

# Decoding Focus: A Prototype Study of Saliency-Conditioned EEG Decoding

Anonymous CVPR submission

Paper ID 21177

## Abstract

001 *Reconstructing visual experiences from brain signals is a*  
002 *central challenge in neural decoding, with growing rele-*  
003 *vance for cognitive science and brain-computer interfaces.*  
004 *While recent models have improved how EEG signals map*  
005 *to visual embeddings, none have yet explored the incorpora-*  
006 *tion of selective attention, i.e. the specific regions of a visual*  
007 *scene that an individual focuses on. We introduce SEEG, a*  
008 *novel EEG-to-image reconstruction framework that incor-*  
009 *porates attention cues by conditioning the EEG encoder*  
010 *on saliency features. These features are extracted from*  
011 *saliency maps generated by DeepGaze, a model trained on*  
012 *human eye-tracking data. Using Feature-wise Linear Mod-*  
013 *ulation (FiLM), we inject this attention information into*  
014 *the EEG encoding process, guiding the model toward more*  
015 *perceptually relevant representations. While this setup as-*  
016 *sumes oracle access to image-derived saliency at test time,*  
017 *it provides an upper bound on how much biologically in-*  
018 *spired attention cues can improve neural decoding. Sys-*  
019 *tematic experiments on the THINGS-EEG2 dataset show*  
020 *that incorporating such priors yields notably sharper and*  
021 *more semantically aligned reconstructions, offering a valu-*  
022 *able benchmark for future EEG-only or predicted-saliency*  
023 *models.*

## 024 1. Introduction

025 Understanding and decoding human visual experience from  
026 brain activity has long been a central goal in neuroscience  
027 and AI. With the rise of deep learning and powerful vision-  
028 language models, recent work has made significant strides  
029 in mapping non-invasive EEG signals to high-level image  
030 embeddings. These methods enable not only classification  
031 and retrieval from neural data but also impressive image re-  
032 constructions.

033 Current EEG-to-image models treat every part of a visual  
034 scene the same, but that doesn't match how our brains ac-  
035 tually work. In reality, visual attention acts like a spotlight,  
036 focusing on important parts of what we see and shaping the  
037 brain's responses [7]. This focused processing modulates

the neural signals measured with EEG, as demonstrated by  
numerous studies showing attention-related changes in both  
early and late visual cortical areas [5, 10]. Yet most exist-  
ing methods ignore this valuable source of information. By  
including visual attention into the EEG-to-image pipeline,  
we can better capture how the brain prioritizes information,  
leading to clearer and more accurate image reconstructions  
that reflect what truly stands out to the viewer.

In this work, we propose SEEG (Saliency-Enhanced  
EEG), a new architecture for EEG-based image reconstruc-  
tion that integrates attention guidance via visual saliency  
maps. These maps, computed directly from the visual stim-  
uli using the DeepGaze IIE saliency predictor [17], are in-  
jected into the EEG encoder using FiLM layers [18]. This  
mechanism allows SEEG to dynamically modulate its inter-  
nal representations based on regions of perceptual impor-  
tance.

Our contributions are as follows:

- We introduce the first saliency-conditioned EEG encoder  
for image reconstruction, using FiLM to modulate fea-  
tures with attention priors.
- We present a systematic comparison of saliency integra-  
tion strategies, FiLM variants, and injection locations,  
providing a practical blueprint for attention-aware neural  
decoding.
- We conduct an upper-bound study, using oracle saliency  
maps to assess the maximum potential benefit of incor-  
porating accurate attention priors into EEG encoding and  
demonstrate that it significantly outperforms the state of  
the art in embedding quality and reconstruction fidelity.

This work is intended as an initial exploration into the  
potential benefit of integrating visual-attention priors into  
EEG encoding. To isolate this effect, we employ saliency  
maps predicted from the ground-truth stimuli, thus provid-  
ing a controlled, oracle-like condition. This setup is not  
intended as a deployable inference pipeline but as a proto-  
type establishing an empirical upper bound on the improve-  
ments achievable if accurate saliency estimates were avail-  
able from EEG or auxiliary modalities.

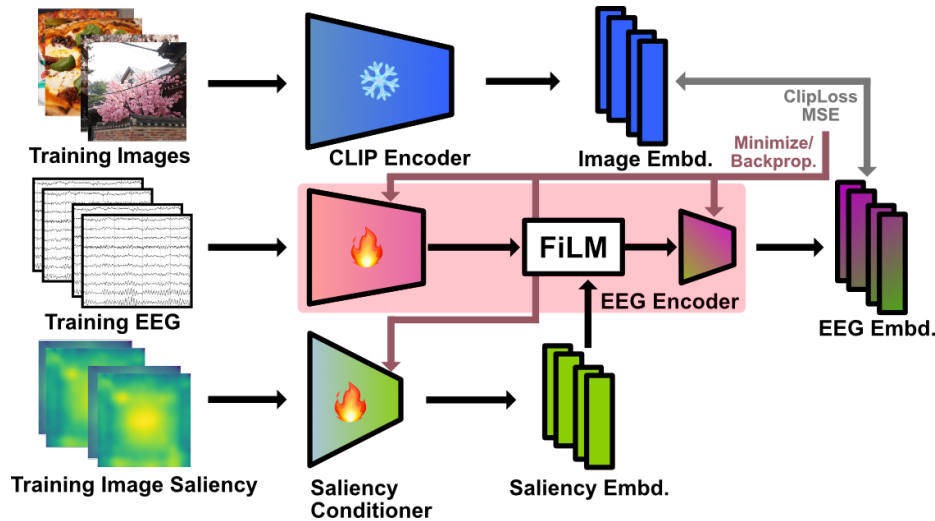


Figure 1. SEEG Architecture: We expand upon the ATM-S framework by injecting saliency directly into the EEG Encoder

## 2. Related Work

Recent advances in generative modeling and multimodal learning have enabled promising progress in reconstructing visual experiences from EEG. At the same time, there is growing interest in integrating biologically inspired mechanisms such as visual attention. We first review methods for EEG-based image reconstruction, then discuss how saliency and attention have been used in AI and neural decoding.

### 2.1. Visual Reconstructions from Neural Activity

The field of neural decoding has experienced a great transformation with the arrival of deep generative models. This is especially true for the analysis and reconstruction of visual imagery from EEG signals. Early works made use of Variational Autoencoders [11, 22] and Generative Adversarial Networks [12, 21] to generate visual content from brain activity. More recently, the focus has turned toward Diffusion Models, which have shown remarkable improvements in reconstruction quality [9, 16, 20].

Most current methods follow a two-stage approach. Firstly, an encoder transforms the EEG signals into latent vectors. Secondly, these latent vectors are used to condition a pre-trained diffusion model, meaning they guide the model’s image generation process to reconstruct visual stimuli that correspond to the brain signals. While Transformer-based encoders have already shown great performance in this first step, there’s still much room for innovation and enhancement in how the EEG vector representations are obtained.

Several methods use multimodal encoders to improve alignment between brain signals and visual outputs. Among them many employ contrastive learning techniques to map EEG and image data into a shared latent space. These mul-

timodal approach, which incorporate images[3] and sometimes also text [20], have proven effective at capturing richer, more nuanced information.

One standout encoder is the Adaptive Thinking Mapper (ATM-S) introduced by Li et al. [16], which is both multimodal and uses the two-stage approach. It is currently the state-of-the-art framework used for EEG-based visual reconstruction. We use ATM-S as a baseline for comparison in this work.

### 2.2. Saliency in AI and Neural Decoding

The most common approach to represent subjects’ visual attention are saliency maps, images which highlight the most attention-grabbing regions of a visual scene [13]. Models like DeepGaze [14], and its improved successor DeepGaze IIE [17] have achieved near-human accuracy in predicting eye fixations, generating highly accurate saliency maps. DeepGaze IIE performed just 0.04% below the human baseline on the MIT/Tübingen benchmark [15]. Due to its high accuracy, we use DeepGaze to generate saliency data for our image set.

Though saliency is widely used in model explainability and training, such as in saliency-guided dropout [4] and data augmentation [2], its role in neural decoding remains limited. Prior work by Khaleghi et al. [12] reconstructed saliency maps and grayscale images from EEG using a GAN, while Delvigne et al. [6] used saliency to enhance EEG-based emotion recognition. However, none of these studies use saliency information to improve the encoder performance itself. Instead they either use them as an intermediary step in the generation pipeline or for feature fusion after training.

In contrast, our work introduces saliency much earlier in the pipeline: directly into the EEG encoder. By incorporat-

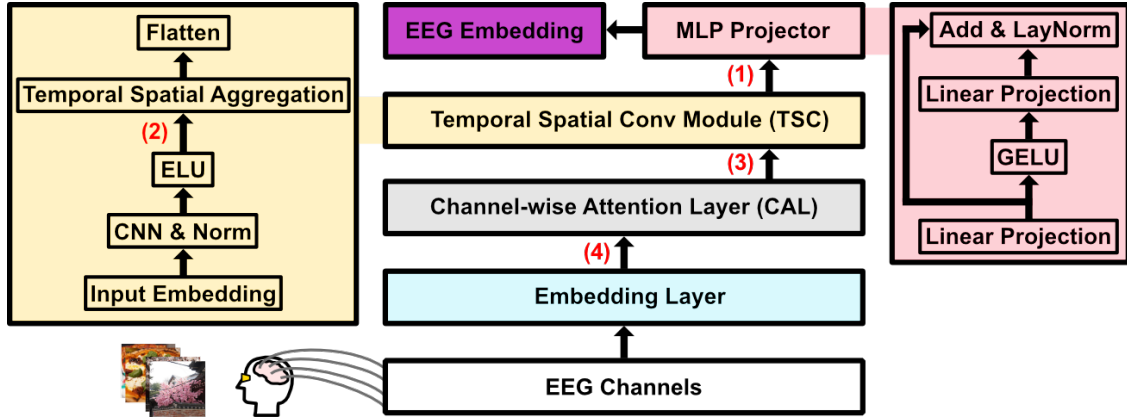


Figure 2. EEG Encoder: Adaptive Thinking Mapper (ATM) [16] with saliency integration locations (1) to (4)

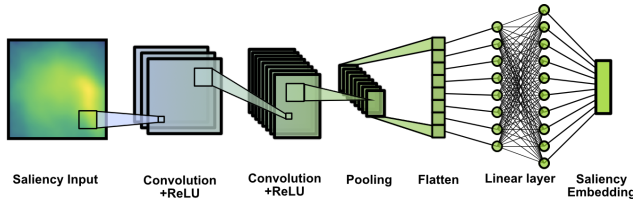


Figure 3. Saliency conditioner architecture overview

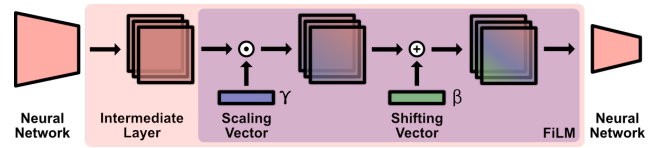


Figure 4. Feature transformation process using FiLM

ing visual attention information at the representation learning stage, we aim to guide the encoder toward more perceptually relevant features before any image reconstruction takes place. To our knowledge, this is the first approach to fuse model-predicted saliency with EEG signals at the embedding level, offering a novel strategy for improving both interpretability and reconstruction fidelity in neural decoding.

### 3. Data

All experiments use the THINGS EEG2 [8] dataset, which includes EEG responses from 10 participants to 1,854 natural images. The training set contains 1,654 categories, each shown four times; the test set includes 200 images from unseen categories, with 80 trials averaged per image to improve signal quality. We report results across all subjects but, like Li et al. [16], we highlight Subject 8

### 4. Method

The SEEG architecture (Figure 1) builds upon the ATM-S pipeline by injecting saliency into the EEG encoder. To do this, saliency maps are generated for all images using DeepGaze IIE.

For both training and testing, saliency maps are computed from the corresponding stimulus images using DeepGaze IIE. This design intentionally provides the model

with oracle saliency to quantify the maximum potential benefit of integrating human-like attention information. In practical scenarios where the ground-truth images are unavailable, these saliency maps could be replaced by estimates predicted from EEG or gaze data.

Once DeepGaze saliency maps are generated, each map is then used as input into our custom Saliency EEG Conditioner.

The Saliency Conditioner is a lightweight network designed to transform 2D saliency maps into compact modulation parameters for the conditioning of the EEG encoder (see Figure 3). It consists of two convolutional layers (with 16 and 32 filters), followed by ReLU activations and an adaptive average pooling layer that reduces spatial dimensions. The resulting 32-dimensional vector is then passed through a multi-layer perceptron block to produce the modulation vectors  $\gamma$  and  $\beta$  used in FiLM. Feature-wise Linear Modulation (FiLM) [18] is a conditioning method that allows external inputs to influence neural networks' internal feature representations. It does so by applying input-dependent scaling ( $\gamma$ ) and shifting ( $\beta$ ) to feature maps following the transformation  $\gamma \cdot x + \beta$  as depicted in Figure 4.

We extend the standard FiLM approach by proposing three novel variants: Half-FiLM (eq. 1), which only applies the scaling parameter; Residual-FiLM (eq. 3), which adds the modulated output back to the original input to promote stable learning; and Hybrid-FiLM (eq. 4), which introduces a learnable parameter  $\alpha$  to interpolate between Residual-

FiLM and the original approach. This flexibility allows the model to determine whether saliency should completely overwrite existing EEG structure or act as a minor correction. Finally, we refer to the original approach as Full-FiLM (eq. 2).

$$F_{\text{half}}(x | \gamma) = \gamma \cdot x \quad (1)$$

$$F_{\text{full}}(x | \gamma, \beta) = \gamma \cdot x + \beta \quad (2)$$

$$F_{\text{res}}(x | \gamma, \beta) = x + (\gamma \cdot x + \beta) \quad (3)$$

$$F_{\text{hyb}}(x | \gamma, \beta) = \alpha \cdot F_{\text{full}} + (1 - \alpha) \cdot F_{\text{res}} \quad (4)$$

To evaluate the most effective strategy for integrating saliency into the EEG encoder, we experiment with all four FiLM variants and compare their performance within the framework.

In addition to testing FiLM variants, we also explore four saliency injection points within the EEG encoder to determine where conditioning is most effective. These integration locations are based on the encoder architecture proposed by Li et al. [16] which serves as the backbone for our experiments.

The encoder consists of three main components:

- **Channel Attention Layer (CAL):** Layer that dynamically re-weights feature maps across EEG channels,
- **Temporal-Spatial Convolution (TSC):** Module that extracts low-level spatiotemporal features from raw EEG signals,
- **Multilayer Perceptron (MLP):** Block that projects the learned representation into the latent space used for image reconstruction.

Saliency information is introduced at four distinct points: before CAL, after CAL, within the TSC module and after the TSC (see Figure 2). Each location corresponds to a different level of abstraction, allowing us to assess whether early or late-stage conditioning leads to more effective saliency-informed encoding for downstream tasks.

Following the saliency-conditioned EEG encoding, we adopt the same architecture as Li et al. [16] for aligning EEG features with the visual domain. Specifically, the output of the EEG encoder is mapped into the CLIP [19] latent space, a high-dimensional embedding space trained to align natural language and visual representations. This alignment allows us to leverage CLIP’s semantically rich image representations to guide image reconstruction from EEG signals. To preserve both semantic alignment and feature fidelity, we train this mapping using a combined loss:

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{\text{CLIP}} + (1 - \lambda) \cdot \mathcal{L}_{\text{MSE}} \quad (5)$$

where we set  $\lambda = 0.9$  in all experiments. The contrastive loss  $\mathcal{L}_{\text{CLIP}}$  encourages alignment with CLIP image embeddings, while the mean squared error  $\mathcal{L}_{\text{MSE}}$  helps maintain

embedding magnitude consistency. We use the *Adam* optimizer [1] with a learning rate of  $1e - 4$ , a batch size of 64, and train for 40 epochs.

Once the EEG and image embeddings are aligned, the high-level CLIP representations are used to condition a pre-trained image generation model, enabling the reconstruction of visual stimuli from brain activity. We use the same evaluation metrics as [16] to assess both EEG embedding and image reconstruction quality.

## 5. Experiments and Results

We detail the findings from three key experiments: a comparison of different conditioning mechanisms, an analysis of optimal saliency injection locations, and a final assessment of the full image reconstruction performance.

The ablation studies were conducted on a single subject (subject 8), selected due to their consistently strong performance across all baseline models. While this choice limits generalizability, it allows for a controlled and focused comparison of model variants under high signal-to-noise conditions. Extending these analyses across more participants is an important direction for future work. Final results are also reported across all subjects.

Because our current setup employs oracle saliency derived from stimulus images, the reported results should be interpreted as an upper bound on achievable performance, rather than as evidence of an end-to-end EEG-only reconstruction capability. Future studies will replace oracle saliency with predicted or generic priors (e.g., center bias, EEG-based saliency prediction) to evaluate robustness under realistic inference conditions.

### 5.1. Comparison of FiLM Conditioning Methods

To evaluate the effectiveness of different FiLM conditioning mechanisms for integrating saliency, we compare the four variants—Half-FiLM, Full-FiLM, Residual-FiLM, and Hybrid-FiLM—against the original baseline architecture without saliency integration. All FiLM-based models use saliency maps generated by DeepGaze IIE, which were exponentiated and normalized based on findings from a preliminary preprocessing study. For consistency, saliency was injected after the TSC block in the EEG encoder for all variants.

To evaluate model performance across experiments, we report final test accuracy, defined as the average test accuracy over the last 30 training epochs, peak accuracy, the highest test accuracy achieved by a model during training and convergence epoch, the epoch at which the model reaches 98% of its final test accuracy.

The results, summarized in Table 1 demonstrate a clear performance benefit from incorporating saliency information via FiLM. All four saliency-conditioned models significantly outperformed the original unconditioned base-



Table 1. EEG Encoder performance comparison of FiLM Variants (best values in bold)

| FiLM Variant  | Final Acc.    | Peak Acc.     | Conv. Epoch | Variance              | Std. Dev.     |
|---------------|---------------|---------------|-------------|-----------------------|---------------|
| Original      | 0.3941        | 0.4750        | 13          | $6.74 \times 10^{-4}$ | 0.0245        |
| Full FiLM     | 0.4381        | 0.4800        | 21          | $5.02 \times 10^{-4}$ | 0.0208        |
| Half FiLM     | 0.4022        | 0.4400        | 26          | $3.50 \times 10^{-4}$ | <b>0.0164</b> |
| Residual FiLM | 0.4437        | <b>0.4950</b> | 29          | $9.50 \times 10^{-4}$ | 0.0274        |
| Hybrid FiLM   | <b>0.4477</b> | 0.4800        | <b>28</b>   | $6.68 \times 10^{-4}$ | 0.0236        |

Table 2. Performance comparison of saliency injection locations (best values in bold).

| Injection Location | Final Acc.    | Peak Acc.     | Conv. Epoch | Variance              | Std. Dev.     |
|--------------------|---------------|---------------|-------------|-----------------------|---------------|
| Original           | 0.3941        | 0.4750        | <b>13</b>   | $6.74 \times 10^{-4}$ | 0.0245        |
| Before CAL         | 0.4180        | 0.4600        | 24          | $4.32 \times 10^{-4}$ | 0.0184        |
| After CAL          | 0.4378        | 0.4750        | 22          | $5.63 \times 10^{-4}$ | 0.0212        |
| In TSC             | 0.4123        | 0.4500        | 26          | $5.30 \times 10^{-4}$ | 0.0205        |
| After TSC          | <b>0.4381</b> | <b>0.4800</b> | 21          | $5.02 \times 10^{-4}$ | <b>0.0208</b> |

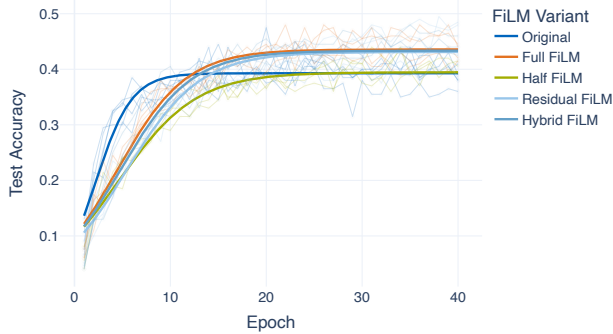


Figure 5. Test Accuracy across epochs shows that while the Original ATMs architecture converged faster, FiLM variants improve over the baseline, most notably Full FiLM, Half FiLM and Hybrid FiLM

line. The highest final accuracy was achieved by the Hybrid FiLM model, with a value of 0.4477, showcasing its ability to dynamically adjust its conditioning based on learned representations. The Residual FiLM model achieved the highest peak accuracy, reaching 0.4950, though it also exhibited the highest variance, suggesting increased sensitivity. In contrast, the Half FiLM variant, while providing only a modest performance gain, proved to be the most stable, with the lowest variance and standard deviation across the different configurations.

The Full FiLM approach showed a good balance. As shown in Figure 5 it delivered comparable performance to the Hybrid and Residual variants, significantly improving over the baseline while maintaining robust, faster and more consistent training. Due to this, we chose to train all models with Full FiLM. However, future work could benefit from incorporating residual and hybrid FiLM since they lead to

performance gains.

## 5.2. Analysis of Saliency Injection Location

Following the success of the FiLM variants, this experiment aimed to identify the most effective location within the EEG encoder to inject the saliency signal. We compared four distinct integration points using the robust Full FiLM variant: immediately after the temporal-spatial convolution (TSC) module, within the TSC module, before the channel attention layer (CAL), and after the CAL. This analysis sought to understand where the saliency information could best guide the encoding process to produce the highest-quality EEG embeddings.

As detailed in Table 2, the injection location had a notable impact on performance. All saliency-enhanced configurations outperformed the original model in terms of final accuracy, further validating the overall approach. The most successful configuration was to inject saliency after the TSC module, which yielded a final test accuracy of 0.4381 and the highest peak accuracy of 0.4800. This configuration also demonstrated the lowest variance among the saliency-conditioned models, indicating a superior balance of performance and stability. This finding supports the hypothesis that saliency, a high-level visual attention signal, is most beneficial when integrated at a point where the EEG features have already been sufficiently abstracted by the initial convolutional layers. While other locations, such as before and after the CAL, also showed improvements, they did not reach the same level of performance and stability as the post-TSC injection.

## 5.3. Impact of Saliency on Image Reconstructions

To evaluate whether the improvements in embedding quality from the previous experiments translated into better image reconstructions, we compared the final visual outputs

Table 3. Quality comparison of image reconstructions from the ATM-S and SEEG model. The values represent the mean of each metric across the 200 test categories. Mean and Standard Deviation (Stdev) are calculated among three distinct reconstructions from one EEG embedding.  $\uparrow$  indicates higher is better,  $\downarrow$  lower is better.

| Encoder       | PixCorr $\uparrow$ | SSIM $\uparrow$ | AN(2) $\uparrow$ | AN(5) $\uparrow$ | Inception $\uparrow$ | CLIP $\uparrow$ | EffNet $\uparrow$ | SwAV $\downarrow$ |
|---------------|--------------------|-----------------|------------------|------------------|----------------------|-----------------|-------------------|-------------------|
| ATM-S (Mean)  | <b>0.1567</b>      | 0.3279          | 0.7462           | 0.8393           | 0.7096               | <b>0.7518</b>   | 0.7104            | 0.6005            |
| ATM-S (Stdev) | 8.7e-3             | 1.4e-3          | 7.8e-3           | 10.5e-3          | 9.8e-3               | 7.5e-3          | 6.9e-3            | 0.6e-3            |
| SEEG (Mean)   | 0.1539             | <b>0.3383</b>   | <b>0.7609</b>    | <b>0.8514</b>    | <b>0.7429</b>        | 0.7282          | <b>0.7169</b>     | <b>0.5985</b>     |
| SEEG (Stdev)  | 3.1e-3             | 1.2e-3          | 1.2e-2           | 1e-2             | 1e-2                 | 8.6e-3          | 1.7e-3            | 1.8e-3            |

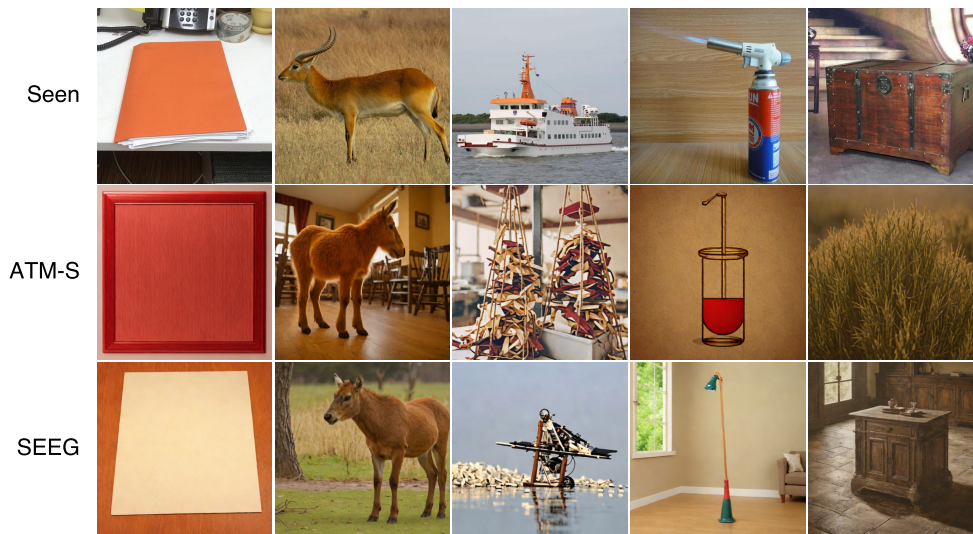


Figure 6. Comparison of EEG-to-image reconstructions. The first row presents the images that were observed during EEG recordings, the second row depicts the reconstructions by the original ATM-S model and the third row shows our Saliency Guided reconstructions. Note the angle of the notebook (column 1), the orientation of the animal (column 2), and the chest (column 3)

of subject 8 on two pipelines: the original ATM-S baseline and our proposed saliency-guided pipeline (SEEG), which used the optimal configuration identified in our studies (Full FiLM conditioning with saliency injected after the TSC module).

The results show that our saliency-conditioned pipeline consistently produces reconstructed images of higher quality. While quantifying the quality of reconstructed images is inherently challenging due to its subjective nature, we utilized a comprehensive set of metrics. As shown in Table 3, our implementation of the baseline model closely replicated the performance of the original work, validating its use as a foundation for comparison. In a direct comparison, the saliency-guided reconstructions demonstrated superior semantic alignment and structural fidelity as can be directly observed in Figure 6.

Visually, the reconstructed images from our pipeline were sharper and contained more semantically plausible features, particularly in areas of high saliency. For example, reconstructions of images with a prominent central object, such as an animal, would render the key features of

that object with greater clarity and detail than the baseline. This qualitative improvement was supported by quantitative gains across various metrics, confirming that guiding the EEG encoder with saliency information significantly enhances the final visual output of the image reconstruction pipeline.

#### 5.4. Results Across Subjects

To evaluate the generalizability of our findings beyond the ablation subject, we assessed the performance of SEEG across all participants in the THINGS EEG2 dataset. These results demonstrate that SEEG consistently outperforms the ATM-S baseline in both embedding quality and downstream classification performance.

On average, SEEG achieved a 4.95% improvement in final test accuracy over ATM-S across all subjects. In addition to this overall accuracy gain, we evaluated N-way classification accuracy, a common metric in neural decoding that measures the model’s ability to correctly identify the ground-truth image from among N candidate images. Specifically, we evaluate results for 2-way, 4-way, and 10-

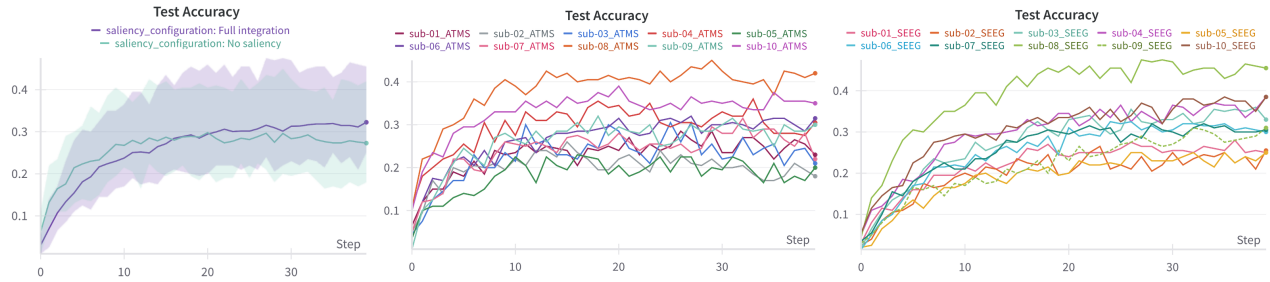


Figure 7. Comparison of SEEG and ATMS Test Accuracies results averaged across all subjects as well as the individual plots of each encoders performance for all subjects

way accuracies, corresponding to increasingly difficult retrieval tasks with 1 correct image and  $N-1$  distractors.

SEEG showed modest improvement in the 2-way setting (+0.4%) but demonstrated more substantial gains in the 4-way and 10-way settings (+3.15% each), indicating that the saliency-guided representations are particularly beneficial in more complex, discriminative scenarios. The relatively small margin in 2-way accuracy suggests that the baseline ATM-S model is already quite reliable when fewer distractor images are present. However, the greater improvements at higher  $N$  values imply that saliency conditioning helps the model better capture semantic distinctions, especially under more ambiguous conditions where perceptual focus is key.

Averaged subject-level results are visualized in Figure 7. While the majority of participants exhibited consistent performance improvements under SEEG, Subject 6 showed a slight decrease in test accuracy compared to the baseline. This minor exception highlights the presence of inter-subject variability, which is common in EEG-based decoding tasks. Despite this, the overall trend remains strongly in favor of saliency conditioning, reinforcing its robustness and broad applicability across individuals.

## 6. Discussion and Conclusion

This work establishes a proof of concept for integrating visual-attention priors into EEG-to-image reconstruction. By conditioning the EEG encoder on oracle saliency derived from the ground-truth stimuli, we estimate the upper performance bound of such attention-guided mechanisms. Although this design involves test-time access to image-based information, it offers clear evidence that attention cues can substantially improve feature quality and reconstruction fidelity.

Our systematic analysis revealed that exponentiating and normalizing raw saliency data is crucial for stable training. Among the FiLM-based conditioning strategies, Hybrid and Residual FiLM delivered the greatest performance gains,

though Full FiLM provided a strong balance of performance and simplicity. The optimal injection location was found to be after the temporal-spatial convolution (TSC) block, supporting the idea that high-level attention signals are most effective when introduced after the EEG input has been abstracted into richer features.

These improvements in embedding quality directly translated into better image reconstructions. SEEG produced images with higher semantic and perceptual similarity, confirming that the additional attention information results in more semantically plausible visual outputs. This work aligns with existing literature and advances the field by bridging the gap between low-level EEG signals and high-level visual semantics using a biologically inspired prior.

While this paper provides a strong proof of concept, it has limitations. The ablation studies were restricted to a single subject, which limits the generalizability of their findings. Additionally, the use of computational saliency as a proxy for real-world attention warrants further investigation. Future research could address these points by confirming the ablation studies results on multiple participants and leveraging real-time eye-tracking data.

Future work will eliminate oracle information by predicting saliency directly from EEG or eye-tracking data, thereby transforming SEEG from a prototype study into a fully realistic decoding framework.

By demonstrating that attention-related signals can enhance image reconstruction from EEG, we have shown a promising direction for improving the interpretability and accuracy of these systems. The methodology presented here lays a solid foundation for developing more robust, context-aware neurotechnologies aligned with human perception.

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